Class-Imbalance Mitigation on BRACS: Replacing Virchow2 with a LoRA-Adapted UNI2-h Encoder

Yannik Queisler

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Abstract

Under a LoRA-adapted Virchow2 encoder, six long-tailed mitigation methods recovered up to about half of the balanced accuracy that BRACS loses when its training classes are imbalanced at a ratio of 100. This report repeats that experiment with UNI2-h as the only change: the same 30 patient-disjoint split-draw shards, class allocations, strength grids, training length, and bootstrap weights, so every UNI2-h estimate is paired with its Virchow2 counterpart. Before any fit, a quantity was declared encoder-dependent only if its paired difference has a 95% interval that excludes zero and a size of at least one point. No damage, recovery, balanced gain, or imbalance-specific recovery meets that rule at any allocation. Under UNI2-h, cross-entropy loses 5.97 points at ratio 100, against 4.95 under Virchow2, a difference of 1.02 points whose interval includes zero. CUDA again gives the largest recovery (2.75 points, share 0.46), closely followed by both forms of GCL and train-time logit adjustment at 2.2 to 2.3 points; logit adjustment and GCL's classifier-only form remain the only methods that substantially lift tail recall, and no method closes the gap. Of six conclusions checked at ratio 100, four carry over and two weaken. CUDA's imbalance-specific recovery becomes material (1.71 points, interval 0.23 to 3.45), although it differs from its Virchow2 value by only 0.51 points, and classifier-only methods recover as much as end-to-end ones only in matched pairs, not as a family. The method ranking at ratio 100 agrees closely between encoders (Spearman 0.86). On BRACS, the measured effects of mitigation under LoRA therefore do not depend on which of the two encoders is adapted, within a resolution of about two points.

1 Motivation

The companion mitigation study on BRACS trained six established long-tailed methods with a Virchow2 encoder (Zimmermann et al., 2024) adapted through rank-8 LoRA updates (Queisler, 2026b). At an imbalance ratio of $\rho = 100$, cross-entropy (CE) lost 4.95 points of balanced accuracy, CUDA recovered about half of that loss, logit adjustment about two-fifths, and no method closed the gap. Net of each method's effect on balanced data, GCL's classifier-only form and cRT corrected more of the imbalance than their recovery against CE showed, and CUDA less. That study named its own scope as its main limitation: one dataset, one encoder, and one adaptation scheme.

A second encoder can shift these results in two ways. It can change the damage that the methods have to recover, through a different balanced accuracy or a different sensitivity to the imbalanced allocation. It can also change how the methods act, because CUDA's image augmentation, GCL's feature clouds, and the classifier-only re-fits all operate on the encoder's representation. Under a frozen linear readout, replacing Virchow2 with UNI2-h (Chen et al., 2024; Mahmood Lab, 2025) left BRACS's imbalance damage unresolved between encoders (Queisler, 2026a), but that comparison did not include adaptation or mitigation. This report asks: *do the LoRA damage on BRACS and each method's recovery depend on the encoder, or do the conclusions drawn under Virchow2 transfer to UNI2-h?*

2 Design

The experiment replicates the Virchow2 mitigation study (Queisler, 2026b) with one change, the encoder. Everything that defines a fit is held fixed: the 30 split-draw shards (three patient-disjoint splits, ten draws each, $G = 10$ training patients per lesion class and a budget of 2,240 patches), the training patients and patches of every draw, the random assignment of classes to allocation ranks, the validation and test patients, the four allocations (balanced, $\rho = 50$, $\rho = 100$, and BRACS's native class shares N), the eight method variants and their strength grids, the training length, and the bootstrap seed. Because the shards and allocations coincide, the two encoders see identical training data, and any difference between them can be attributed to the encoder together with its interaction with the fixed recipe.

UNI2-h enters in its pinned public release (Mahmood Lab, 2025), the same checkpoint used for the frozen comparison (Queisler, 2026a). The patch embedding is UNI2-h's own pooled token (1,536 dimensions), whereas Virchow2 concatenates its class token with the mean patch token (2,560 dimensions); each encoder thus keeps the pooling its authors recommend. The adaptation is the same for both: a rank-8 LoRA update ($\alpha = 16$) on the query-key-value and output projections of every attention block, trained with a linear head (a cosine head for GCL, with scale $s = 30$ and a second-stage sampler of effective-number range $a = 0.9999$, $b = 10^{-5}$). The 600 end-to-end training steps were tuned on CE under Virchow2 at $\rho = 100$ and were deliberately not re-tuned, so that the training recipe is identical across encoders; the cost of this choice is discussed in Section 5. Across the 30 shards the experiment runs 1,170 end-to-end and 1,230 classifier-only fits, and every configured fit completed and entered the analysis.

3 Evaluation

Each encoder is first analysed with the endpoints of the Virchow2 study. The primary endpoint is test patient-macro balanced accuracy (BA). For method m with validation-selected strength m^* and allocation a, the CE damage, the recovery, its share of the damage, the balanced gain, and the imbalance-specific recovery are

$$\begin{gathered} D_a = \mathrm{BA}(\mathrm{CE}, 1) - \mathrm{BA}(\mathrm{CE}, a), \qquad R_m = \mathrm{BA}(m^*, a) - \mathrm{BA}(\mathrm{CE}, a), \\ G_m = \mathrm{BA}(m^*, 1) - \mathrm{BA}(\mathrm{CE}, 1), \qquad R_m - G_m, \end{gathered} \tag{1}$$

with $\mathrm{share}_m = R_m / D_a$ computed per replicate. A recovery is *material* if it is at least one point and its interval lies above zero, and *full* if the interval of $\mathrm{BA}(m^*, a)$ contains the point estimate of $\mathrm{BA}(\mathrm{CE}, 1)$. Balanced sampling and both forms of logit adjustment reduce to CE on the balanced allocation, so $G_m = 0$ for them by construction. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates, and intervals are 95 percent percentile intervals.

The encoder contrast was fixed in the configuration before any fit. For each quantity X among $\mathrm{BA}(\mathrm{CE}, 1)$, D_a, R_m, G_m, and $R_m - G_m$,

$$\delta X = X(\text{UNI2-h}) - X(\text{Virchow2}), \tag{2}$$

computed on the same replicate weights for both encoders, so every δ is paired. A quantity counts as *encoder-dependent* only if the interval of δX excludes zero and $|\delta X| \geq 1$ point. Two descriptive summaries accompany the rule: the agreement of the material-recovery verdict per method, and the Spearman rank correlation of R_m across the eight variants per allocation. Six conclusions of the Virchow2 study were also listed in advance to be checked at $\rho = 100$: (i) CUDA gives the largest recovery; (ii) logit adjustment and classifier-only GCL are the only methods that substantially lift tail recall; (iii) no share interval reaches 1; (iv) CUDA's imbalance-specific recovery is not material, whereas those of classifier-only GCL and cRT are; (v) classifier-only

methods recover about as much as end-to-end ones; and (vi) the LoRA model on the balanced allocation is less accurate than the frozen readout.

The frozen UNI2-h readout lost $D_{100} = 7.19$ points $[5.23, 9.32]$ from a balanced BA of 38.23 percent (Queisler, 2026a). That study used a different set of draws, so the comparison with it is descriptive and has no paired interval.

4 Results

The results first describe the UNI2-h experiment on its own, with the endpoints of the Virchow2 study, and then compare the two encoders through the pre-fixed paired contrast and the six listed conclusions.

4.1 Damage under UNI2-h

The balanced CE model reaches 36.08 percent BA $[33.11, 39.08]$, 0.45 points below its Virchow2 counterpart. It loses $D = 4.32$ points $[2.48, 6.17]$ at $\rho = 50$ and 5.97 points $[4.13, 7.97]$ at $\rho = 100$; at native shares the damage of 0.72 points $[-0.46, 1.91]$ cannot be distinguished from zero. All three damages lie within about one point of the Virchow2 values (3.54, 4.95, and 0.85). As under Virchow2, the LoRA damage at $\rho = 100$ is smaller than the frozen readout's 7.19 points, and again because of the balanced model: LoRA's 36.08 percent stays 2.15 points below the frozen UNI2-h readout's 38.23 percent, while the imbalanced LoRA model (30.11 percent) lands within one point of the frozen value implied by its damage (31.04 percent). These frozen values come from different draws and are descriptive.

4.2 Recovery per method

Figure 1 and Table 1 give each method's paired recovery under UNI2-h. At $\rho = 100$, six of eight variants recover a material part of the damage, the same six as under Virchow2: CUDA (2.75 points $[1.47, 4.12]$), end-to-end and classifier-only GCL (2.34 and 2.24), train-time and post-hoc logit adjustment (2.28 and 2.04), and balanced sampling (2.03). cRT (0.74) and DisAlign (0.23) again recover a positive amount below the one-point threshold. CUDA's share of the damage, 0.46 $[0.26, 0.71]$, is the largest, followed by end-to-end GCL (0.39), train-time logit adjustment (0.38), and classifier-only GCL (0.37). No share interval reaches 1, and at $\rho = 100$ no method formally reaches full recovery: CUDA's accuracy interval, $[30.45, 35.42]$, ends below the balanced 36.08 percent.

At $\rho = 50$, CUDA (2.05 points, share 0.47), end-to-end and classifier-only GCL (1.96 and 1.90), post-hoc logit adjustment (1.50), and balanced sampling (1.22, interval $[0.09, 2.42]$) recover a material part. Train-time logit adjustment gains 1.09 points with an interval that just touches zero, $[-0.01, 2.25]$. CUDA and both GCL forms formally reach full recovery at $\rho = 50$, but their paired shares, with intervals ending below 0.80, show that the gap is not closed. At native shares, seven of eight recoveries lie between -1.13 and 0.36 points with intervals that include zero. CUDA gains 0.95 points $[0.09, 1.91]$, just below the one-point threshold, and reaches 36.31 percent, above the balanced CE model.

4.3 Imbalance-specific recovery

The balanced gains repeat the Virchow2 pattern with smaller or equal magnitudes (Table 2). CUDA improves the balanced model by $G = 1.04$ points, with an interval that now just includes zero, $[-0.04, 1.96]$. cRT loses 0.53 points $[-1.06, -0.02]$, the two GCL forms lose 0.36 and 0.79 points with intervals that include zero, and DisAlign is indistinguishable from CE.

Net of these gains, classifier-only GCL corrects 3.03 points $[1.27, 4.76]$ of the damage at $\rho = 100$, a share of 0.51 $[0.26, 0.70]$, and end-to-end GCL 2.69 points; both are again the

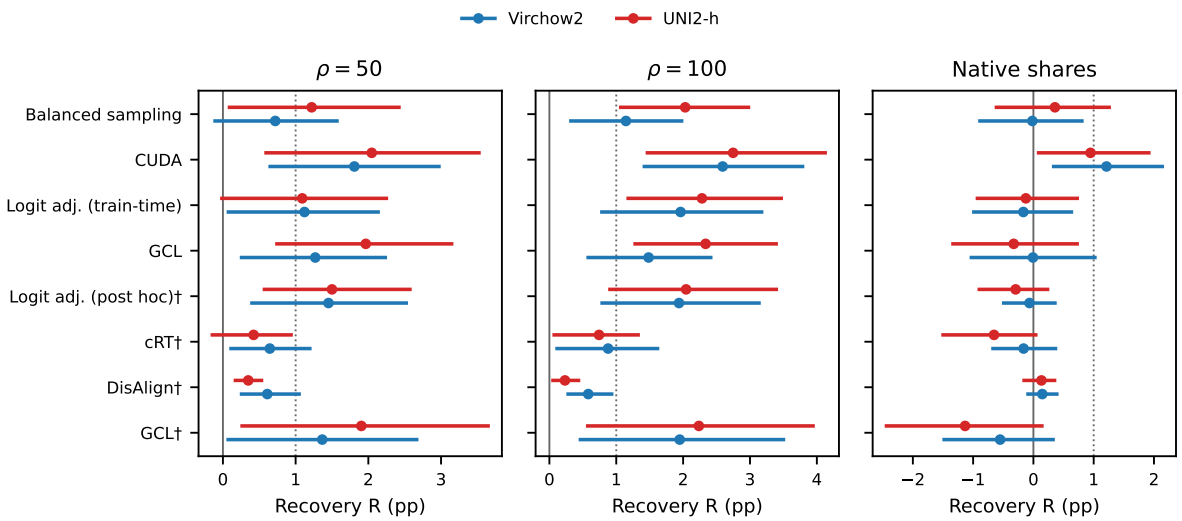


Figure 1: Under both encoders, CUDA, GCL, and logit adjustment recover one to three points at $\rho = 50$ and $\rho = 100$, and at native shares no method except CUDA moves. Paired recovery R_m with 95% intervals for Virchow2 (blue) and UNI2-h (red); the solid line marks zero and the dotted line the one-point threshold. †Classifier only.

largest imbalance-specific recoveries. cRT's 1.28 points $[0.38, 2.09]$ remain material. CUDA's imbalance-specific recovery, 1.71 points $[0.23, 3.45]$ or a share of 0.29, becomes material under UNI2-h, whereas under Virchow2 its interval included zero (1.20 points $[-0.16, 2.62]$). At $\rho = 50$ only the two GCL forms are material (2.70 and 2.32 points); CUDA (1.01, interval $[-0.59, 2.87]$) and cRT (0.96) fall short. At native shares every imbalance-specific recovery lies within half a point of zero.

4.4 Encoder contrast

No quantity meets the pre-fixed rule for encoder dependence at any allocation. Table 3 lists the paired contrasts at $\rho = 100$, and Figure 2 shows δR_m for all three imbalanced allocations. The largest point contrast at $\rho = 100$ is the damage itself: UNI2-h loses 1.02 points more than Virchow2, but the interval, $[-0.29, 2.48]$, includes zero. Five of the eight recoveries are larger under UNI2-h in point estimate, by up to 0.89 points (balanced sampling) and 0.85 points (end-to-end GCL), which is consistent with a somewhat larger damage to recover; none of these intervals excludes zero. The only intervals that exclude zero belong to DisAlign, whose recovery and imbalance-specific recovery are 0.35 and 0.34 points smaller under UNI2-h, well below the one-point threshold. The balanced gains differ by at most 0.36 points. At $\rho = 50$ and at native shares no $|\delta|$ exceeds 0.82 points, and no interval excludes zero.

The descriptive summaries point the same way under imbalance. At $\rho = 100$ all eight material-recovery verdicts agree between encoders, and the Spearman correlation of R_m across the eight variants is 0.86. At $\rho = 50$ the correlation is 0.88 and six of eight verdicts agree; the two that differ sit at the threshold, balanced sampling (material only under UNI2-h, lower bound 0.09) and train-time logit adjustment (material only under Virchow2, lower bound -0.01 under UNI2-h). At native shares the correlation is 0.71 and seven verdicts agree, the exception being CUDA (1.21 points under Virchow2, 0.95 under UNI2-h). There, all recoveries but CUDA's are indistinguishable from zero under both encoders, so their ranking is dominated by noise.

Method	$\rho = 50$ ($D = 4.32$)		$\rho = 100$ ($D = 5.97$)		N ($D = 0.72$)
	R_m	share	R_m	share	R_m
CUDA	**2.05** [0.59, 3.52]	0.47 [0.17, 0.79]	**2.75** [1.47, 4.12]	0.46 [0.26, 0.71]	0.95 [0.09, 1.91]
GCL	**1.96** [0.74, 3.15]	0.46 [0.24, 0.69]	**2.34** [1.28, 3.39]	0.39 [0.23, 0.60]	−0.33 [−1.33, 0.73]
Logit adj. (train-time)	1.09 [−0.01, 2.25]	0.25 [−0.00, 0.44]	**2.28** [1.18, 3.47]	0.38 [0.23, 0.55]	−0.13 [−0.93, 0.73]
GCL†	**1.90** [0.26, 3.65]	0.44 [0.08, 0.71]	**2.24** [0.57, 3.94]	0.37 [0.12, 0.57]	−1.13 [−2.44, 0.14]
Logit adj. (post hoc)†	**1.50** [0.57, 2.57]	0.35 [0.17, 0.53]	**2.04** [0.91, 3.39]	0.34 [0.19, 0.49]	−0.29 [−0.90, 0.23]
Balanced sampling	**1.22** [0.09, 2.42]	0.28 [0.03, 0.49]	**2.03** [1.07, 2.98]	0.34 [0.20, 0.49]	0.36 [−0.61, 1.26]
cRT†	0.42 [−0.15, 0.94]	0.10 [−0.04, 0.21]	0.74 [0.07, 1.33]	0.12 [0.01, 0.21]	−0.65 [−1.50, 0.04]
DisAlign†	0.35 [0.17, 0.53]	0.08 [0.04, 0.14]	0.23 [0.05, 0.44]	0.04 [0.01, 0.07]	0.13 [−0.15, 0.35]

Table 1: Under UNI2-h, six variants recover at least one point at $\rho = 100$, and no share interval reaches the full damage. Recovery R_m in BA points and share of the CE damage, with paired 95% intervals; bold marks material recovery ($R_m \geq 1$, interval above zero). †Classifier only, frozen encoder; unmarked rows train end to end. Shares at native shares are omitted because the damage there cannot be distinguished from zero.

Method	G_m	$R_m - G_m$			$\rho = 100$
		$\rho = 50$	$\rho = 100$	N	share
CUDA	1.04 [−0.04, 1.96]	1.01 [−0.59, 2.87]	**1.71** [0.23, 3.45]	−0.09 [−1.31, 1.29]	0.29 [0.04, 0.54]
GCL	−0.36 [−1.27, 0.51]	**2.32** [1.05, 3.71]	**2.69** [1.43, 4.00]	0.03 [−1.17, 1.32]	0.45 [0.25, 0.71]
GCL†	−0.79 [−1.77, 0.17]	**2.70** [0.86, 4.76]	**3.03** [1.27, 4.76]	−0.34 [−1.98, 1.18]	0.51 [0.26, 0.70]
cRT†	−0.53 [−1.06, −0.02]	0.96 [0.15, 1.67]	**1.28** [0.38, 2.09]	−0.12 [−1.23, 0.78]	0.21 [0.07, 0.34]
DisAlign†	−0.01 [−0.06, 0.03]	0.36 [0.17, 0.56]	0.25 [0.06, 0.45]	0.15 [−0.14, 0.39]	0.04 [0.01, 0.08]

Table 2: Under UNI2-h, classifier-only GCL again corrects the largest part of the imbalance, and CUDA's imbalance-specific recovery is material at $\rho = 100$. Balanced gain G_m and imbalance-specific recovery $R_m - G_m$ (Eq. 1) in BA points, with its share of the CE damage at $\rho = 100$; paired 95% intervals, bold marks material recovery. †Classifier only. Balanced sampling and logit adjustment have $G_m = 0$ by construction.

4.5 Conclusions of the Virchow2 study

Table 4 checks the six conclusions listed in advance. Four carry over to UNI2-h. CUDA again gives the largest recovery at $\rho = 100$, although its lead over the next method shrinks from 0.63 to 0.41 points, and differences between methods are not tested. Logit adjustment and classifier-only GCL remain the only methods that substantially lift tail recall (Section 4.6). No share interval reaches 1, and the LoRA model on the balanced allocation stays below the frozen readout. Two conclusions weaken. CUDA's imbalance-specific recovery is material under UNI2-h; classifier-only GCL and cRT remain material, so the conclusion fails only in its CUDA part. Its contrast with Virchow2, 0.51 points [−1.34, 2.52], is not resolved, so the verdict changes because an estimate near the threshold moved, not because the encoders demonstrably differ. Classifier-only methods recover as much as end-to-end ones within the matched pairs: post-hoc logit adjustment trails its train-time form by 0.24 points, and classifier-only GCL trails end-to-end GCL by 0.10 points. As a family they recover less: excluding CUDA, the mean share at $\rho = 100$ is 0.37 for the end-to-end and 0.22 for the classifier-only methods, against 0.31 and 0.27 under Virchow2. The gap widens because balanced sampling gains under UNI2-h while DisAlign and cRT recover little. These family and pair comparisons are point estimates without intervals.

Quantity	Method	Virchow2	UNI2-h	δ (UNI2-h − Virchow2)
BA(CE, 1)	–	36.53	36.08	-0.45 $[-2.13, 1.21]$
D_{100}	–	4.95	5.97	1.02 $[-0.29, 2.48]$
R_m	CUDA	2.59	2.75	0.16 $[-1.13, 1.63]$
	Logit adj. (train-time)	1.96	2.28	0.32 $[-0.89, 1.56]$
	Logit adj. (post hoc)†	1.94	2.04	0.11 $[-0.67, 0.96]$
	GCL	1.48	2.34	0.85 $[-0.32, 2.09]$
	GCL†	1.95	2.24	0.29 $[-1.12, 1.71]$
	Balanced sampling	1.14	2.03	0.89 $[-0.17, 1.93]$
	cRT†	0.88	0.74	-0.14 $[-0.92, 0.55]$
	DisAlign†	0.58	0.23	-0.35 $[-0.67, -0.08]$
G_m	CUDA	1.39	1.04	-0.36 $[-1.93, 1.12]$
	GCL	-0.33	-0.36	-0.03 $[-1.19, 1.23]$
	GCL†	-0.64	-0.79	-0.15 $[-1.48, 1.31]$
	cRT†	-0.48	-0.53	-0.06 $[-0.71, 0.65]$
	DisAlign†	-0.01	-0.01	-0.00 $[-0.06, 0.06]$
$R_m - G_m$	CUDA	1.20	1.71	0.51 $[-1.34, 2.52]$
	GCL	1.81	2.69	0.88 $[-0.67, 2.60]$
	GCL†	2.59	3.03	0.44 $[-1.49, 2.26]$
	cRT†	1.36	1.28	-0.08 $[-1.28, 0.92]$
	DisAlign†	0.59	0.25	-0.34 $[-0.67, -0.07]$

Table 3: No paired encoder contrast at $\rho = 100$ excludes zero with a size of at least one point. Point estimates per encoder in BA points and the paired contrast δ (Eq. 2) with its 95% interval. †Classifier only.

4.6 Where recovery lands

Under UNI2-h, the CE model at $\rho = 100$ keeps 2.7 percent recall in the tail third of classes by assigned rank, against 33.9 percent when balanced, and gains 16.6 points in the head (Figure 3). Logit adjustment raises the tail to 15.3 (train-time) and 19.0 percent (post hoc), at a cost of six to nine points of head recall. Classifier-only GCL raises it to 24.9 percent; its head recall, 35.4 percent, equals the balanced model's 35.2, and its body recall, 35.2 percent, is 4.4 points below the balanced model's. CUDA and end-to-end GCL recover through the body instead, 38.2 and 37.0 against 25.1 percent, and raise the tail only to 9.3 and 7.6 percent. Balanced sampling, cRT, and DisAlign leave the tail between 3.7 and 7.9 percent. Every method thus places its recovery in the same rank group as under Virchow2, while the tail recalls of CE, logit adjustment, and classifier-only GCL are two to five points lower under UNI2-h. Rank-group recalls are point estimates.

4.7 Selected strengths and probability quality

Validation selection behaves as under Virchow2 for most methods. At $\rho = 100$, train-time logit adjustment selects the full correction $\tau = 1$ in 23 of 30 shards and post-hoc logit adjustment in 15, balanced sampling selects $s = 1$ in 23, and GCL selects $\sigma = 0.05$ in 18 (end to end) and 16 (classifier only) of 30. CUDA selects $p_{\text{aug}} = 1$ in 14 of 30 shards at $\rho = 100$ and in 12 at $\rho = 50$, never at native shares, and never on the balanced allocation, where 0.5 wins 18 shards and 0.25 wins 12. DisAlign is the exception. Under UNI2-h it selects the largest value of its grid, $\gamma = 2$, in 12 of 30 shards at $\rho = 100$, and $\gamma = 3$, the largest value of the wider native-shares grid, in 19 of 30 shards at native shares. Under Virchow2, $\gamma = 1$ or 1.5 won 22 of 30 shards at $\rho = 100$. DisAlign's grid was copied unchanged by design, so under UNI2-h it may stop short of DisAlign's best strength.

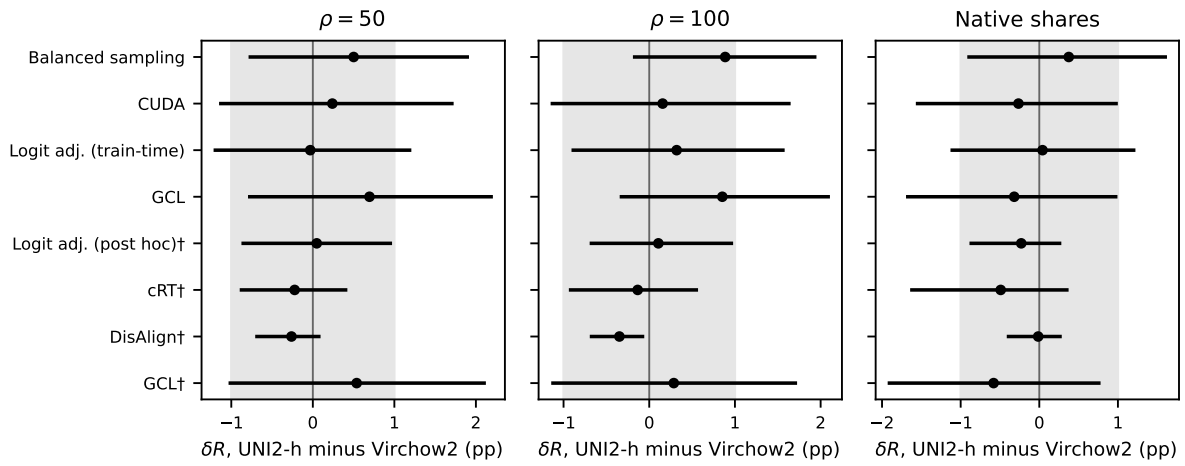


Figure 2: Every paired recovery contrast overlaps zero or stays inside the one-point band. Paired contrast δR_m (UNI2-h minus Virchow2) with 95% intervals; the grey band marks ±1 point. †Classifier only.

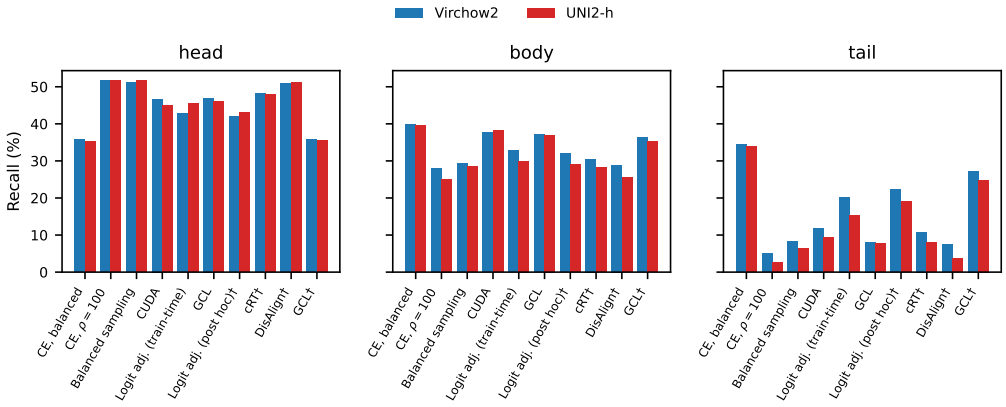


Figure 3: Under both encoders, only logit adjustment and classifier-only GCL lift tail recall substantially at $\rho = 100$, and CUDA and end-to-end GCL gain in the body. Patient-macro recall of the head, body, and tail thirds of classes by assigned rank, averaged over shards, for Virchow2 (blue) and UNI2-h (red). †Classifier only.

Imbalance again mainly worsens raw calibration. At $\rho = 100$ the CE model's raw negative log-likelihood (NLL) rises from 2.95 to 3.58 nats and its raw expected calibration error (ECE) from 31.5 to 39.8 points. CUDA has the lowest raw NLL, 2.26 nats $[2.10, 2.43]$, and raw ECE, 21.5 points $[17.8, 26.7]$, below even the balanced CE model; the two logit-adjustment forms follow (3.00 and 2.89 nats), and cRT (4.04) and end-to-end GCL (3.91) are highest. After validation temperature scaling, every method lies between 1.75 (CUDA) and 1.80 nats, around CE's 1.80 and close to the balanced 1.66, and scaled ECE ranges from 10.4 (balanced sampling) to 14.3 points (DisAlign) against CE's 14.1. As under Virchow2, CUDA curbs raw overconfidence, but no method improves probability quality beyond what temperature scaling already provides.

5 Discussion

Does mitigation under LoRA depend on the encoder? Not measurably. None of the 20 pre-fixed contrasts per allocation meets the encoder-dependence rule, the material-recovery

Conclusion at $\rho = 100$	Virchow2	UNI2-h	Verdict
(i) CUDA gives the largest R_m	2.59; next 1.96	2.75; next 2.34 (GCL)	Holds
(ii) Only logit adjustment and GCL† lift tail recall substantially	Tail 5.1 to 20.2, 22.3, and 27.3; others $\leq$11.9	Tail 2.7 to 15.3, 19.0, and 24.9; others $\leq$9.3	Holds
(iii) No share interval reaches 1	Highest upper bound 0.82	Highest upper bound 0.71	Holds
(iv) CUDA's $R_m - G_m$ not material; GCL† and cRT† material	1.20 $[-0.16, 2.62]$; 2.59 and 1.36 material	1.71 $[0.23, 3.45]$; 3.03 and 1.28 material	Fails for CUDA
(v) Classifier-only recovers as much as end-to-end	Mean share without CUDA 0.31 vs. 0.27	Mean share without CUDA 0.37 vs. 0.22; pairs within 0.24	Pairs only
(vi) LoRA balanced model below frozen readout	36.53 vs. 39.57	36.08 vs. 38.23	Holds

Table 4: Four of the six pre-listed conclusions carry over to UNI2-h, and the two that weaken do so without a resolved encoder contrast. Values in BA points, recall points, or shares; tail recall is the patient-macro recall of the rarest third of classes for CE, the two logit-adjustment forms, and classifier-only GCL. The frozen readouts in (vi) come from different draws and are descriptive. †Classifier only.

verdicts agree for all eight methods at $\rho = 100$, and the method ranking is stable at both imbalance ratios. The two encoders also share the structure of the result: CUDA, GCL, and logit adjustment recover one to three points at $\rho = 100$; GCL's classifier-only form corrects the most imbalance net of its balanced effect; logit adjustment and that form restore the tail; no method closes the gap; and at native shares nothing but CUDA's general gain moves. The conclusions of the Virchow2 study (Queisler, 2026b) therefore do not depend on that choice of encoder.

What do the two weakened conclusions mean? Both reflect estimates near a threshold, not a resolved difference. CUDA's imbalance-specific recovery rises from 1.20 to 1.71 points and crosses into materiality, but the paired contrast of 0.51 points has an interval almost four points wide. Read across both encoders, CUDA's imbalance-specific part is probably positive and smaller than its recovery against CE, with its balanced gain explaining a third to a half of that recovery; whether it reaches one point is not settled by either experiment alone. The family comparison shifts because three methods moved by less than a point in opposite directions, while the matched pairs, which isolate the effect of adapting the encoder from the method itself, agree within 0.5 points under both encoders. The finding that a frozen-encoder classifier re-fit suffices for a given method holds; the family-level summary does not, because it averages over methods of very different strength.

How large a difference could have been missed? The paired intervals of the recovery contrasts span between about 0.6 and 3.0 points, those of the damage up to 3.2 points, and those of the imbalance-specific recoveries up to 4.2 points. An encoder difference of one point in a single method's recovery would therefore often remain unresolved, and the null result bounds encoder effects at roughly one to two points, not at zero. The point estimates do lean one way: UNI2-h's damage is one point larger, and five of eight recoveries are larger with it. A larger damage that the methods recover in proportion would produce exactly this pattern, and the shares, which differ between encoders by about 0.1 or less, support that reading.

Does UNI2-h change the frozen-versus-LoRA picture? No. Under both encoders, LoRA's balanced model is two to three points less accurate than the frozen readout, and the

smaller LoRA damage reflects this weaker reference rather than a more robust imbalanced model. With UNI2-h, the best mitigated model at $\rho = 100$, CUDA at 32.86 percent, stays 5.4 points below the frozen balanced readout, the same margin as under Virchow2. For BRACS, adapting either encoder with this recipe and then mitigating does not outperform a frozen readout trained on balanced data.

Limitations. The study adds a second encoder but keeps one dataset and one adaptation scheme. All settings were tuned under Virchow2 and transferred unchanged: the 600 training steps, the LoRA rank and target modules, and the strength grids. This keeps the comparison paired, but it measures each encoder under a recipe fitted to the other, and UNI2-h's lower balanced accuracy may partly reflect that. DisAlign's selections concentrate on the largest grid value under UNI2-h, so its small recovery, and the only contrasts whose intervals exclude zero, may partly reflect a truncated grid. The two encoders also differ in their pooled embedding (1,536 against 2,560 dimensions), so the contrast describes two complete encoder-plus-pooling pipelines. The comparisons with the frozen readouts use different draws and are descriptive. Imbalance-specific recovery assumes that a method's balanced effect carries over additively to the imbalanced allocations, as in the Virchow2 study.

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